

BRYAN NGUYEN
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EDUCATION

Virginia Tech, Blacksburg, VA
B.S in Computer Engineering
College of Engineering

Expected Graduation: *May 2028*

GPA: 3.9/4.0

WORK EXPERIENCE

Virginia Tech – SuperDARN HF Radar Lab

Blacksburg, VA

Software Developer Intern

March 2026 – Present

- Engineered an end-to-end data delivery platform and REST API serving **5 TB** of scientific radar data across **4** radar systems, implementing token-based authentication, rate limiting, usage telemetry, role-based access controls, and secure file delivery.
- Improved data-processing throughput by **60%** and reduced processing failures by **50%** for compute and I/O-intensive pipelines by implementing multiprocessing, process isolation, read timeouts, caching, and automated data-generation workflows.
- Built and optimized **6+** scientific visualization pipelines used by **90+ users** to analyze over **60 TB** of radar data, integrating server-side caching, automated parameter selection, coordinate transformations, metadata overlays, and interactive visualization interfaces.

Amazon

Arlington, VA

Extern

May 2025

- Enhanced an internal Auto Scaling scheduling tool by implementing timezone-aware scaling events using TypeScript and date-fns-tz, improving accuracy of multi-region scheduling.
- Developed reusable AWS CDK constructs for CloudWatch alarms, metrics, and resource tagging, standardizing monitoring and cost-management practices across services.
- Streamlined infrastructure operations by automating scaling event configuration, timezone conversions, and retrieval of stack resources directly from CloudFormation outputs.

PROJECTS

Resy Booking Bot

June 2026

- Developed a Dockerized full-stack reservation automation platform using FastAPI, Next.js, TypeScript, and SQLite, integrating external APIs to process booking requests across **500+** venues.
- Engineered persistent background job workflows supporting **10+** concurrent reservation monitors with **5-second** polling intervals, implementing scheduled execution, cancellation, venue resolution, slot filtering, and fault-tolerant API error handling.
- Built an authenticated web console for querying real-time availability, launching and managing reservation jobs, and monitoring **10+** live and historical booking attempts with sub-second status updates.

Storefront

December 2025

- Built a full-stack e-commerce platform with Next.js and TypeScript, using Prisma + PostgreSQL with migrations and seed tooling for reliable data modeling.
- Implemented JWT cookie authentication with hashed passwords and role-based access control, securing admin routes and dashboards.
- Developed server-side product management APIs and a persistent client-side cart with checkout flow, improving UX and system security.

SKILLS

Technical: Java, Python, JavaScript, React, Node.js, PostgreSQL, Git, FastAPI, Docker, GitHub Actions

Languages: Fluent in English, Vietnamese